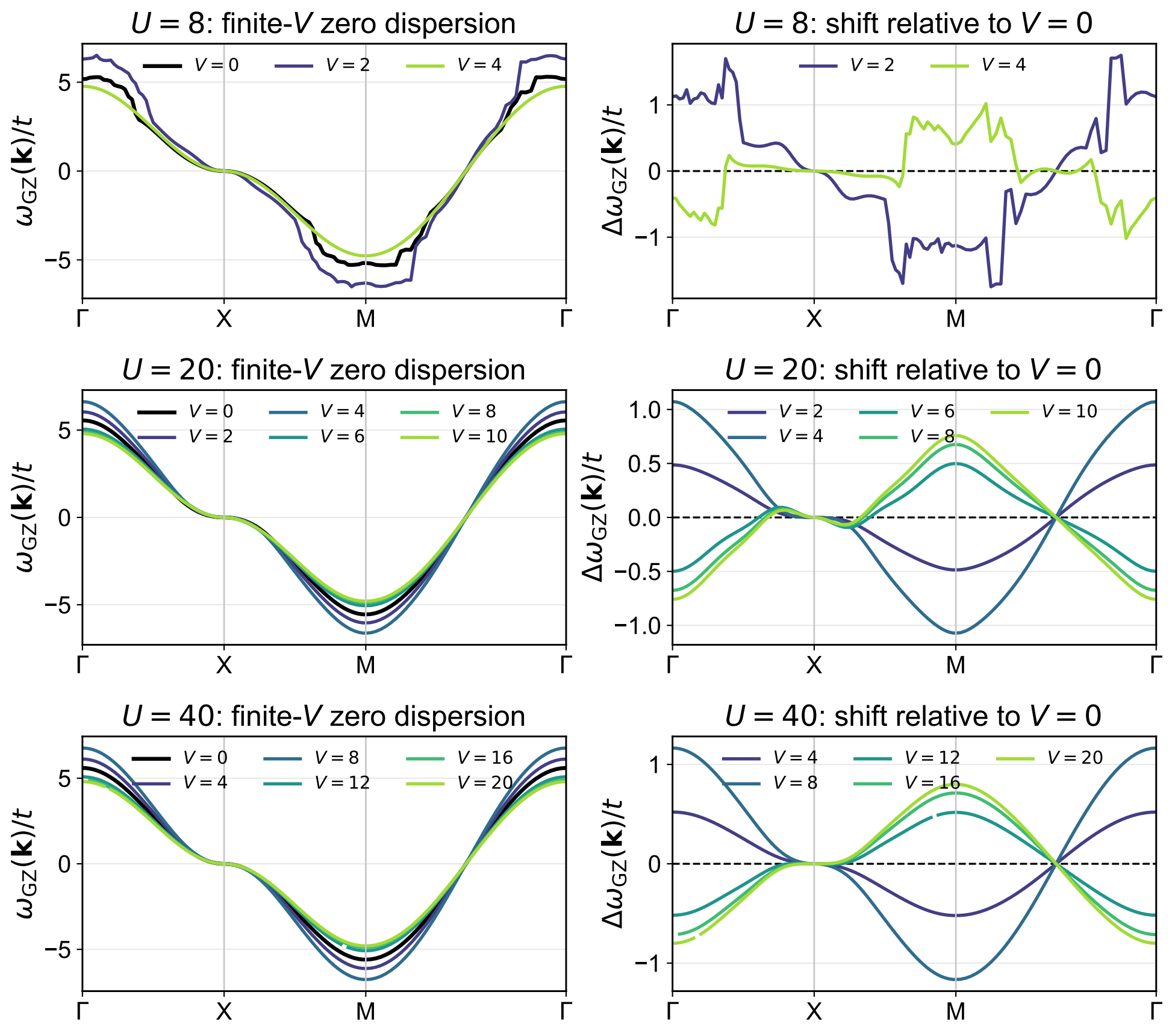
$U = 8$: finite-V zero dispersion
$V = 0$
$V = 2$
$V = 4$
$\omega_{GZ}(\mathbf{k})/t$
5
0
−5
Γ
X
M
Γ
$U = 8$: shift relative to $V = 0$
$V = 2$
$V = 4$
$\Delta\omega_{GZ}(\mathbf{k})/t$
1
0
−1
Γ
X
M
Γ
$U = 20$: finite-V zero dispersion
$V = 0$
$V = 2$
$V = 4$
$V = 6$
$V = 8$
$V = 10$
$\omega_{GZ}(\mathbf{k})/t$
5
0
−5
Γ
X
M
Γ
$U = 20$: shift relative to $V = 0$
$V = 2$
$V = 4$
$V = 6$
$V = 8$
$V = 10$
$\Delta\omega_{GZ}(\mathbf{k})/t$
1.0
0.5
0.0
−0.5
−1.0
Γ
X
M
Γ
$U = 40$: finite-V zero dispersion
$V = 0$
$V = 4$
$V = 8$
$V = 12$
$V = 16$
$V = 20$
$\omega_{GZ}(\mathbf{k})/t$
5
0
−5
Γ
X
M
Γ
$U = 40$: shift relative to $V = 0$
$V = 4$
$V = 8$
$V = 12$
$V = 16$
$V = 20$
$\Delta\omega_{GZ}(\mathbf{k})/t$
1
0
−1
Γ
X
M
Γ